جامعة خليفة
Khalifa University

HAZARDOUS MATERIALS MANAGEMENT

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1. Purpose:

The purpose of this procedure is to outline detailed and practical advice about ways to manage hazardous substances and the environmental, health and safety risks that can be associated with the using, handling, storage and containment of any hazardous substances.

2. Scope:

This procedure is applicable to all personnel who may purchase, use, store and dispose of hazardous materials, including, but not limited to, employees, contractors, subcontractors, visitors, suppliers, and the public, while visiting, working with or on behalf of.

3. Responsibilities:

3.1 EHS Steering Committee Representatives: Responsible for

- Ensure necessary resources are available for the effective implementation and maintenance of this procedure.
- Report to top management on the performance of this procedure and need for improvement.

3.2 Division Managers: Responsible for

- Ensure that this procedure is implemented and maintained in their area of responsibility.

3.3 EHS Manager: Responsible for

- Overall responsibility of the implementation, monitoring and effectiveness of this procedure.
- Communicate and ensure the awareness this procedure to all levels of KU.
- Review the lessons learned from the implementation of this procedure.
- Ensure that all training requirements related to this procedure is identified through coordination with Division Managers / Section Managers / Management System Committee Coordinators.

3.4 EHS Coordinators & Laboratory Engineers: Responsible for

- Effective implementation of this procedure in their area of responsibility.
- Identifying, evaluating, and calculating risk levels for all health and safety risks and environmental impacts related to hazardous materials, moreover, the completion of the EHS Risk Register within their area of responsibility.
- Implementing and maintaining all required operational control procedures within their area of responsibility.

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- Implementing all action plans related to hazardous materials within their area of responsibility.

3.5 EHS Section: Responsible for

- Establish all action plans related to hazardous materials.
- Develop periodical incident reports from hazardous materials management.
- Conducting inspections of work areas handling and hazardous materials to monitor compliance of employees.
- Provide the required training to all concerned.
- Disseminate informative materials regarding chemical handling.

3.6 EHS Inspectors: Responsible for

- Conducting regular housekeeping audits on work areas and chemical storage areas.
- Close monitoring & performance of this procedure.
- Assist all supervisors and employees concerned with the management of hazardous materials.
- Report all incidents and near misses.
- Notify the training requirements for supervisors and employees to EHS System Development and Auditing Section Head.

4. Abbreviation / Definitions:

Substance: Any chemical element and its compounds and biological entity or microorganism, except radioactive substances or genetically modified organisms.

MSDS: Material Safety data Sheet is a form provided by the supplier of chemical product / manufacturer which containing data regarding the properties of a particular substance. It has intended to provide workers and emergency personnel with procedures for handling or working with that substance in a safe manner.

Chemical: Is a substance that produced or used in a process (reaction) involving changes to atoms or molecules. The term sometimes defined more broadly as "a substance".

Health hazard: Means a chemical for which there is statistically significant evidence based on at least one study conducted in accordance with established scientific principles that acute or chronic health effects may occur in exposed employees. The term "health hazard" includes chemicals, which

are carcinogens, toxic or highly toxic agents, reproductive toxins, irritants, corrosives, sensitizers, hepatotoxins, nephrotoxins, neurotoxins, agents that act on the hematopoietic system, and agents which damage the lungs, skin, eyes, or mucous membranes.

Physical hazard: Means a chemical for which there is scientific valid evidence that it is a combustible liquid, a compressed gas, explosive, flammable, an organic peroxide, an oxidizer, pyrophoric, unstable (reactive) or water-reactive.

Mitigation: Measures taken to reduce the consequences of a potential hazardous event. The limitation of undesirable effects of a particular event.

5. Procedure:

5.1 General:

The ability of hazardous substances to cause adverse health effects lies largely with the physical properties of the substance, along with the dose/ amount of exposure that is likely to occur and by which entry routes into the body, exposure may occur.

5.1.1 Knowledge of hazards:

- Know the hazards associated with the materials you are using. Review labels and MSDS sheets.
- Know emergency procedures.
- Know the locations of safety equipment such as emergency shower, eyewash, fire extinguisher, fire alarm, and emergency phone numbers.

5.1.2 Personal practices:

- Do not eat, drink, smoke, chew gum, or apply cosmetics, lip salve, contact lenses in areas where chemicals are used or stored.
- Do not store food items or cosmetics in areas where chemicals are used or stored.
- Confine long hair and loose clothing when working with chemicals.
- Wear shoes with closed toes and closed heels.
- Wear appropriate personal protection equipment including gloves and eye protection.
- Do not smell or taste chemicals.

- Do not use mouth suction for Pipetting.
- Always wash hands and other exposed skin after chemical use.

5.1.3 Using chemicals:

- Date and sign chemicals when received and when opened.
- Ensure incoming chemicals are on the inventory list
- For chemicals that degrade in quality or become unsafe after prolonged storage also, date them with the shelf-life expiration date.
- When using a chemical for the first time read the label carefully and consults the MSDS.
- Ensure that all required safety equipment is available and in good working order, including materials and equipment for spill response.
- Conduct a visual inspection of the container and its contents routinely.
- Ensure that all containers are in good condition and properly labeled.

5.1.4 Control of hazards:

- Use appropriate personal protective equipment.
- Conduct all processes that may result in the release of toxic vapors, fumes or dust within the fume hood or other adequate containment device.
- Do not leave potentially hazardous chemical processes unattended.
- Handle and store glassware with care to avoid damage and dispose of any damaged glassware in an appropriate sharps container.
- Always rinse used glassware and plastic ware after use and before giving to wash-up staff or placing in the garbage.
- Wear appropriate respiratory equipment when air contaminant concentrations cannot be sufficiently restricted by engineering controls.

5.1.5 Storage of chemicals:

- Store all hazardous chemicals in a secure location, accessible only to authorized laboratory workers.
- To store large containers of hazardous liquids securely near the floor but at a height, that allows safe ergonomic handling.
- Minimize quantities of chemicals kept in the work area.

- Store chemicals under appropriate conditions: At appropriate temperature and humidity levels, away from heat sources such as steam pipes or ovens, dry and adequately vented storage locations, away from direct sunlight, electric motor, on shelves and in cabinets that are secured to prevent tipping.
- Do not store liquid chemicals above eye level.
- Use only explosion-proof refrigerators and freezers for storage of flammable liquids.
- Do not store unsegregated chemicals alphabetically. Segregate chemicals by hazard class (e.g., flammable liquids, organic acids, oxidizers, reactive chemicals) and store separately.

5.1.6 Health Effects:

Hazardous substances may cause short term (acute) and / or longer term (chronic) health effects. It is therefore important that we become familiar with the physical properties of the substances that we use here at work.

5.1.7 Routes of Exposure

There are five main ways that we can or may be exposed to hazardous substances. These are

- Inhalation – Breathing in of contaminants.
- Ingestion – Where substances are eaten or swallowed.
- Absorption – Of substances into our system through the skin.
- Injection – Where a substance is actually forced into our body e.g. with a syringe or due to extremely high pressure.
- Cuts and Open wounds – the substance can enter the body through damaged or broken skin.

It is important when using any hazardous substance that we become familiar with their physical properties and routes of exposure. This helps to develop and establish what risks exist and what controls or precautions / actions can be taken when working with a specific hazardous substance, to prevent it causing us harm.

Manages hazardous materials through specific indicators and objectives that cover the following:

- Air quality (fumes initiated, filtration efficiency)

- Water Quality (effects of accomplished tasks or daily testing and analysis operations)
- Land Management (waste of utilized materials, chemical, biological residues)

Generally, a hazardous material is one, which may pose a risk to the health of anyone coming into contact with it in the workplace. The contact may be in handling, storage, transport or disposal of the materials.

Chemicals and other hazardous materials are present in 's everyday work, from the cleaning chemicals being used to the hazardous materials used in food testing and analysis, which may result in fumes, wastes or spills, chemical, or biological kits also pose a risk to the health and safety of persons in the workplace to some extent.

All department/sections, whose work includes the use or management of hazardous materials, establish operational control procedures and ensure preventative and correction actions are set accordingly.

5.2 Classification of Hazardous Materials:

5.2.1 Hazardous Materials Management Indicators:

- **Class 1** Indicators are associated with the health, safety, and well-being of the public.
- **Class 1A** indicators relate to general provisions for hazardous materials management.
- **Class 1B** indicators relate to specific provisions for hazardous materials management.

- **Class 1A Indicators and Objectives – General Provisions for Hazardous Materials Management:**

o All hazardous materials are classified according to the Globally Harmonized System for Classification and Labeling of Chemicals (GHS). Hazardous materials determined to be hazardous waste are being classified according to the Abu Dhabi Emirate Waste Classification System as detailed in the Code of Practice – Waste Management and managed accordingly.

- Proper management of hazardous materials is the responsibility of employees, suppliers, and contractors involved with the hazardous materials' use. Roles and responsibilities of key groups are defined according to the type of hazardous material and as stated within AD EHSMS CoP 01 – Hazardous Materials.

- **Class 1B Indicators and Objectives Specific Provisions for Hazardous Substances Management:**

- The provisions of this clause apply to hazardous substances and mixtures.
- The provisions of this clause do not apply to the management of substances identified as hazardous waste.
- Hazardous substances determined to be hazardous waste are being managed according to AD EHSMS CoP 54 – Waste Management.
- Materials and mixtures are assigned hazard classes based on the harmonized classification criteria for materials, substances and mixtures in the following list for specific hazard classes or a group of closely related hazard classes:

1. Physical hazards:

- Explosives;
- Flammable gases;
- Flammable aerosols;
- Oxidizing gases;
- Flammable liquids;
- Flammable solids;
- Self-reactive substances and mixtures;
- Pyrophoric liquids;
- Pyrophoric solids;
- Self-heating substances and mixtures;
- Substances and mixtures, in contact with water, emit flammable gases;
- Oxidizing liquids;
- Oxidizing solids;

- Organic peroxides; and
- Corrosive to metals.

2. Health hazards:

- Acute toxicity;
- Skin corrosion / irritation;
- Serious eye damage/eye irritation;
- Respiratory or skin sensitization;
- Germ cell mutagenicity;
- Carcinogenicity;
- Reproductive toxicity;
- Specific target organ systemic toxicity – single exposure;
- Specific target organ systemic toxicity – repeated exposure; and
- Aspiration hazard.

3. Environmental hazards:

- Hazardous to aquatic environment.

5.3 Hazard Communication:

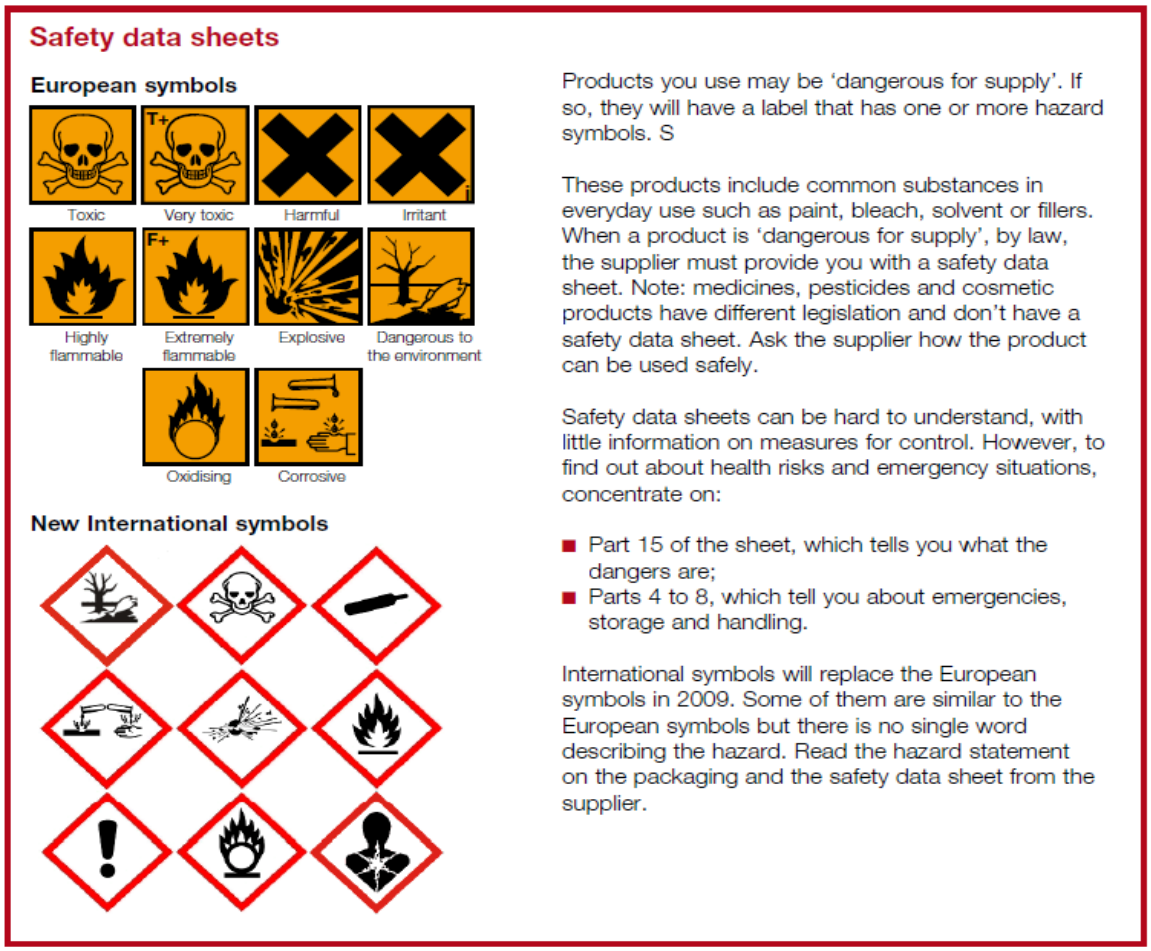
All divisions and sections ensure clear communication about their hazardous materials. An inventory of all hazardous materials being kept with the Management System Committee Coordinators related to the activities of their divisions / sections. For each item listed, the following information and action are required:

- For each chemical in the Division / Section, the inventory identifies the chemical name, part number or Chemical Abstract Service Number, quantity, location and type of hazard (carcinogenic, acute toxicity and reproductive toxicity).
- Name & address of manufacturer and supplier with contact numbers, if possible
- Each Department/Section will maintain a MSDS sheet for each chemical listed in the inventory.
- The MSDS sheet will not have a revision date older than 3 years.
- The Management System Committee Coordinators will update the computerized inventory list using the Chemical Inventory List posted at its location.

- Physically verify that all chemicals on the list are present.
- Visually inspect chemicals for signs of deterioration and past expiry dates.
- Remove all expired or unfit chemicals.
- Details of shipping and storage requirements clearly printed. Labeling employ, such method as is conducive to the storage of the item to ensure that the labeling does not deteriorate.
- Main hazards are clearly indicated – flammable, toxic, corrosive etc.
- Systems or processes to ensure that new chemical are appropriately communicated.
- Systems or processes to ensure that new employees are appropriately made conversant with the procedures involving hazardous materials.
- Transport procedures.

Notes: **The MSDS includes the following information:**

1. Identification of the substance
2. Composition-Chemical components – common names of hazardous material components that compose 1% more of the product (0.1% for carcinogens) / Manufacturer information–contact number included.
3. Hazard identification-Physical hazards–potential for fire, explosion reactivity / Health hazards–those health hazards associated with the product or its components. To include signs and symptoms of exposure and any medical conditions that may be affected by exposure.
4. First aid.
5. Firefighting.
6. Accidental release
7. Handling and storage - Procedures for safe handling.
8. Exposure controls / personal protection- Control practices – engineering controls, work practices & PPE requirement.
9. Physical and chemical properties.
10. Stability and reactivity.
11. Toxicological.
12. Ecological.
13. Disposal.
14. Transport.
15. Regulatory.
16. Other-Date of MSDS preparation or last revision.



5.4 Labeling, Packaging and Hazard Warning:

All divisions / sections will ensure proper labeling and packaging of hazardous materials is done as follows:

- Upon receiving new chemicals record, date of receiving on the bottle as well as name of supervisor.
- Determine appropriate storage location using MSDS.
- Add any new chemicals to the Chemical Inventory List.
- File MSDS in the binder.
- The designated person will periodically update computerized chemical Inventory List and print and post a new list when warranted.
- Substances are classified into hazard categories (broadly, the same as represented by hazard symbols – see Figure 1).

- Material Safety Data Sheets are provided by the suppliers of hazardous materials.
- Packaging is suitable for the material it contains, and;

a. Able to withstand normal handling.

b. Made of material resistant to the effects of its contents.

c. Able to retain its contents if fitted with a replaceable closure.

- Labeling must be clearly shown in indelible marking, securely fixed and include:

a. The name, address and telephone number of the supplier.

b. The name of the substance, both chemical and trade.

c. Hazard symbol(s).

d. Risk information.

e. Safety information.

f. The UN or EEC number (if any).

g. The container of any hazardous substance to be transported by road, sea or air must have fixed to it the appropriate hazard warning diamond.

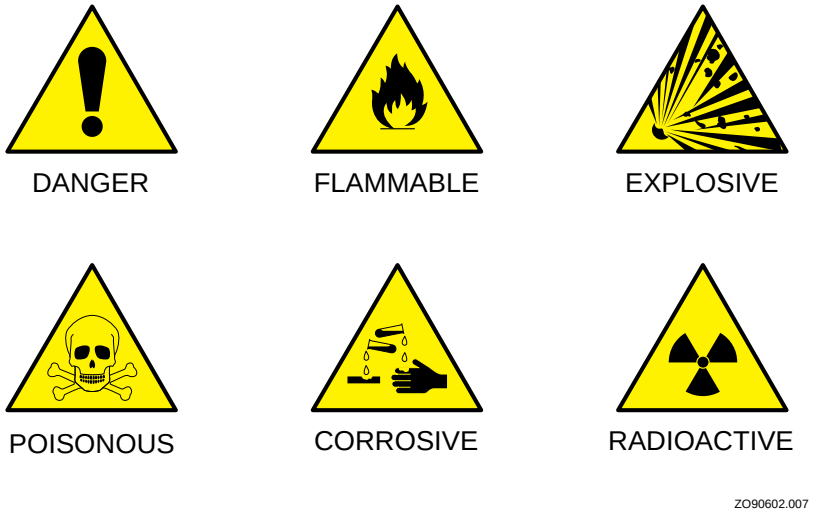


Figure1Hazard Symbols

5.5 Exposure control measures:

- Control measures are always a mixture of equipment and ways of working to reduce exposure. The right combination is crucial. No measures, however practical, can work unless they are used properly.
- Therefore, any 'standard operating procedure' should combine the right equipment with the right way of working. This means instructing, training and supervising the workers doing the tasks.
- You need control measures that work and continue to work – all day, every day.

5.5.1 Choosing control measures:

In order of priority:

a. Eliminate the use of a harmful product or substance and use a safer one.

b. Use a safer form of the product, e.g. paste rather than powder.

c. Change the process to emit less of the substance.

d. Enclose the process so that the product does not escape.

e. Extract emissions of the substance near the source.

f. Have as few workers in harm's way as possible.

g. Provide personal protective equipment (PPE) such as gloves, coveralls and a respirator. PPE must fit the wearer.

If your control measures include 5, 6 and 7, make sure they all work together.

5.5.2 Control equipment:

Control equipment comes in many forms. It includes ventilation to extract dust, mist and fume; glove boxes and fume cupboards; spray booths and refuges (clean rooms in dirty work areas). It also includes using water to reduce dust, and systems for disinfecting cooling water.

For control equipment, your supplier should provide a 'user manual'. If you do not have one, ask for it. Moreover, if this is impossible, you may need professional help to write one. The user manual should set out schedules for checks, maintenance and parts replacement. For example, it should include:

- A description of the system;

- The daily checks the worker or supervisor needs to carry out, e.g. the ventilation turns on, the airflow meter gives the right reading;
- The weekly or monthly checks the supervisor or owner needs to carry out, e.g. of equipment wear and tear, and shortcuts that are not creating dangerous;
- Details of any thorough examination and test;
- Signs of wear and control failure;
- A list of replaceable parts;
- A description of how operators should use the system so it works effectively.

Remedy defects in good time. It is pointless making checks if you take no action when something is wrong. And you are not managing health and safety properly if the 'thorough examination and test' produces a long list of 'actions needed'.

Keep simple records of your checks and actions, e.g. in a log book, and keep these records for at least five years.

5.5.3 Staying in control: checking and maintaining:

Once you have control, you need to keep it. As the employer, you must make sure that the control measures (equipment and the way of working) keep working properly.

You should name someone to be in charge of checking and maintaining control measures. It could be you, or someone you appoint, as long as they know what they need to do, and are able to do it. That is, they are 'competent' to:

- Check that the process isn't emitting uncontrolled contaminants;
- Check that the control equipment continues to work as it was designed;
- Check that workers follow the right way of working.

Two of the most common control measures where maintenance is critical are local exhaust ventilation (LEV) and personal protective equipment (PPE).

5.5.3.1 Local exhaust ventilation (LEV):

If you use local exhaust ventilation to control exposure, it needs regular checking and thorough examination and testing at least once a year.

Many people, e.g. engineers or insurance companies can carry out a thorough examination and testing of LEV.

5.5.3.2 Personal protective equipment (PPE):

Personal protective equipment is often used as part of control measures. This also needs checking and maintenance because, if it fails, it no longer provides protection and exposes the wearer of danger. The users need to know exactly what they are doing, and so do the supervisors.

5.5.4 Skills and experience:

- **Competence**
  Ensure that whoever designs, installs, maintains and tests your control measures is competent – they have the necessary skills, knowledge and experience. You can assess the competence of equipment and service providers with questions such as:
  - Have you done this sort of work before?
  - What are your qualifications?
  - Do you belong to a professional organization?
  - Can I speak to previous clients?
- **Worker involvement**
  Involve staff in developing control measures to make sure they are suitable for the way they carry out the work. Encourage them to suggest improvements, and to report anything they think might be going wrong.
- **Training, instruction and information**
  a. Explain to your workers and anyone else who needs to know, what the dangers are. It has been a poor practice just to hand them a page of written information.
  b. Show workers how to use control measures properly, and how to check that they are working.
  c. Carry out practice drills for cleaning up spills safely – do this before any spillages happen.

d. If workers need to use respirators, they also need face fitting and training.

e. If they need to use protective gloves, they need to know how to put them on and take them off without contaminating their skin. See 'Find out more'.

5.6 Ordering of Chemicals:

- Information on safe handling, storage, and disposal of chemicals should be completely known by the user before making an order.
- Specific information is specified on the purchase order such as size of packaging, required shelf life of material and any other requirements like MSDS or other additional information to be included in the delivery.
- Small packaging is requested to avoid dispensing from bulk containers that should result to spillage during transfer of chemicals.

5.7 Inspection of Delivered Chemicals:

- Stores Division personnel inspect the deliveries of chemicals. Employees who ordered chemicals are asked to inspect the materials received.
- Delivered chemicals must have proper labels, hazard information and MSDS. Packaging without identifying labels and those that are damaged or unsealed will not be accepted.

5.8 Hazardous Materials Handling and Storage:

Safe handling and storage procedures are based on MSDS and manufacturers and / or suppliers' information. The inventories of hazardous goods are readily available for inspection by any employee involved in the handling or storage on hazardous materials.

Hazardous Materials Storage:

5.8.1 Flammable or Combustible Materials:

- Flammable and fireproof storage cabinets are required for flammable or combustible materials. Only such materials may be stored in these cabinets.
- Flammable goods are stored at least 6 meters from oxidizers and other reactive materials.
- Bulk storage has a secondary spill wall or disk at least 1.5 meters high.

- Electrical wirings are explosion proof and storage containers are grounded (earthed) to limit risk from static electrical buildup.
- One-hour fire resistant walls are preferred in storage facilities.
- Adequate signage shall be provided e.g., no smoking/naked lights
- Suitable and sufficient firefighting equipment shall be provided.

5.8.2 Corrosive Materials:

- Corrosives are stored at least 6 meters from other reactive materials.
- Alkalis are stored 6 meters away from acid storage area.
- Corrosive materials are stored in appropriate storage cabinets.
- Eyewash stations with at least 15 minutes of continuous water flow are provided.

5.8.3 Toxic Materials:

- Toxic materials are securely stored in access-controlled cabinets.
- Storage ensures no contamination from the toxic materials to the surrounding environments.
- Eyewash stations with at least 15 minutes of continuous water flow are provided.

5.8.4 Compressed and liquefied gasses and gas cylinders handling and usage:

- Compressed gases are being handled only by competent and properly trained persons. Training shall include information regarding;
- The contents of this guideline as well as any specific information relevant to the gas being used and emergency information
- Safety shoes are required when moving cylinders.
- Cylinders are not being dragged or physically carried.
- Transport cylinders with a hand-truck designed for the transport of cylinders. Cylinder caps secure during transport.
- Prevent damage to cylinders. Locate cylinders where they are being protected from physical damage by striking or falling objects, corrosion or damage from public tampering.
- No person other than the gas supplier attempts to mix gases in a cylinder.

- Cylinders are not being subjected to artificially create low temperatures without approval from the supplier.
- Containers are not being used for any other purpose than holding the contents as received.
- Damaged or leaking cylinders are being reported to the EHS Section Manager immediately for proper disposal.
- Cylinders are not to be picked up by the cap.
- Ropes, chains and slings are not to be used to suspend cylinders, unless cylinder was designed for such.
- Leaking, defective, fire burned and corroded containers are not to be shipped without the approval of the supplier.
- Cylinders should be stored in the correct manner.
- Incompatible cylinders shall not be stored together.
- The storage area should be suitable for the purpose and not exposed to any direct source of heat, e.g. the Sun.
- Adequate signage shall be provided.

5.8.5 General Requirements for Storing Hazardous Materials:

- When feasible automated and enclosed handling is being employed in preference to open handling
- Storage of hazardous materials is in accordance with manufacturer's recommendation to optimize product quality
- Hazardous materials are released for use on the basis of FIFO (First In First Out) to allow them to be used within their shelf life
- Storage takes into account the limitations of packing and containers to ensure that these are not damaged.
- Control of occupational exposure limits for hazardous materials are being maintained
- Adequate ventilation of the storage area is ensured.

5.8.6 General Requirements for Handling Hazardous Materials:

- Employees use non-hazardous materials wherever possible.
- Employees refer to the MSDS and guidelines for the safe use of the hazardous materials.

- Employees conduct Risk assessment when using new hazardous materials.
- Employees open containers with caution in case of deterioration or pressurization.
- Hazardous Materials used in areas with good ventilation.
- Fire extinguishers (carbon dioxide or dry powder) compatible with the hazardous materials used are readily available.
- Personal Protective Equipment are being used all the time when handling hazardous materials.
- Hazardous materials are not to be carried in open containers.
- Employees prevent splashing on skin or clothing while using hazardous materials.
- A register is prepared and available to all personnel, detailing all hazardous materials used on site:

  a. Details of all hazardous materials used in the workplace

  b. The product's Material Safety Data Sheet (MSDS)

  c. The limits of exposure to the hazardous material

  d. The details of any risk assessments carried out

  e. The details of any incidents or near misses as a result of being exposed to hazardous materials at the facility.

5.8.7 Exposure to Hazardous Materials:

In case of exposure to any hazardous material, the following steps shall be followed:

- Isolate employees and others from the source of the exposure.
- Render first aid. Refer to the Material Safety Data Sheet for emergency treatment details and if necessary seek medical advice.
- Deal with the source of exposure in accordance with guidelines on the MSDS – contact emergency services if necessary.
- Ensure that exposure does not reoccur.
- Complete procedural steps in accordance with the Incident/ Accident Reporting and Investigation Procedure.
- Review procedures.

5.9 Employee Training:

Section Heads / Management System Committee Coordinators, shall identify all training requirements related to the handling and storage of hazardous materials and ensure that EHS personnel / external sources as applicable provide necessary training.

Training programs are developed through the knowledge of the procedures, processes, systems, or materials to be used, and appropriate training content is developed through such means as:

- Reviewing' accident, near miss and injury records to identify how accidents occur and what can be done to prevent them from recurring.
- Requesting employees to provide, in writing and in their own words, descriptions of their jobs. These should include the tasks performed and the tools, materials and equipment used.
- Observing employees at the worksite as they perform tasks, asking about the work, and recording their answers.
- Examining similar training programs offered by other similar organizations.

The employees themselves can provide valuable information on the training they need. Safety and health hazards and environmental aspects are identified through the employees' responses to such questions as whether anything about their jobs frightens them, if they have had any near-miss incidents, if they feel they are taking risks, or if they believe that their jobs involve hazardous operations or substances.

Suggested EHS General Training:

5.9.1 Hazardous Materials:

- Flammable and Combustible Liquids / Gasses.
- Explosives and Blasting Agents.
- Bulk Delivery and Mixing Vehicles.
- Storage and Handling of Compressed and Liquefied Gases.
- Process Safety Management of Highly Hazardous Chemicals.
- Contractor Responsibilities.
- Hazardous Waste Operations and Emergency Response.
- Hazardous Waste Cleanup Workers.

- New Technology Programs.
- Personal Protective Equipment.
- General Environmental Controls.
- Medical Services and First Aid.

5.9.2 Fire Protection

- Fire Protection / Prevention.
- Fire Risk Assessment.
- Training and Education on Basic Fire Chemistry.
- Portable Fire Extinguishers.
- Fixed Extinguishing Systems.
- Fire Detection and Alarm Systems.

5.9.3 Materials Handling and Storage.

5.9.4 Electrical Safety-Related Work Practices.

5.9.5 Toxic and Hazardous Substances.

5.9.6 Toxic, Hazardous, and Radioactive Substances.

- Ionizing/ non ionizing Radiation Safety.
- Hazard Identification.
- Occupational Exposure to Hazardous Chemicals in Laboratories.

5.10 Documentation:

Documentation requirements of Hazardous Materials Management include MSDS, Chemical Inventory List, Records and Registers. These documents are to be made available when necessary for employees, public authorities and the emergency services. Section Managers/ Area Supervisors are ensuring that records of hazardous materials used in their work areas are maintained, and are readily available to all personnel.

5.11 Internal Notification and External Reporting:

- If an employee discovers a spillage or leaking container in a workplace, they must raise the alarm, contact the relevant Section Head / Supervisor, and advise them of the situation. Once advised the Section Head / area Supervisor set in place the emergency procedures as

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detailed in the MSDS. Reporting is completed in line with Near Miss Reporting Procedure and Incident Reporting and Investigation Procedure.

- KU ensures that EHS reporting to the Abu Dhabi EHSMS Centre is based on accurate data monitoring and collection. KU provides periodic reports on the use of Hazardous Materials and their discharges to air, water and land. This is divided into two main categories:

  a. Total discharge rate – The rate at which an entire discharge enters the environment, expressed as mass or volume per unit time, for example in units of m3/hour for emissions, liter / second for effluents or kg / year for wastes; and

  b. Pollutant discharge rate – The rate at which an individual specified pollutant within a discharge enters the environment, expressed (preferably) as the "mass flow rate" in units of mass per unit time, e.g. mg/ second, g/hour or kg/ year or, (failing that) as "volume flow rate" in units of volume per unit time e.g. liter / second, m3 / year.

- KU's EHS reporting is also part of the corporate culture and is part of the 'corporate social responsibility' reports that measure the social and environmental performance along with the operational performance.

- As well as producing yearly reports, where required, there may be instances where KU has to report to the Abu Dhabi EHSMS Centre more frequently. For example, in the case of new hazardous materials importing and use, hazardous material exceeding permit limits for emissions to air, water or land or in the case of accidental spill or major incidents. For details of reports, refer to KU Performance Monitoring and Reporting Procedure.

5.12 Exposure monitoring:

Monitoring normally means air sampling but it may also involve taking biological samples, e.g. breath or urine. Monitoring normally refers to 'Workplace Exposure Limits' (WELs) published by HSE. These limits should not be exceeded.

5.13 Emergency Management:

Emergency preparedness and response related to Hazardous Materials will be instigated in line with the Emergency Management Procedure.

6. Reference Standards:

6.1 Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF) - CoP-01 Hazardous Material.

6.2 ISO 45001:2018 Occupational health and safety management systems – Requirements. 4.4.6.
6.3 ISO14001:2015 Environmental Management System – Requirements. 4.4.6.

7. Annexure:

7.1 Annexure 1 – Nil

8. Records / Documents / Formats:

8.1 Chemical Inventory List Form No.: KU – FD – EHS – FOR – 73.
8.2 Hazardous Materials Register No.: KU – FD – EHS – REG – 23.
8.3 Hazardous Materials Register Form No.: KU – FD – EHS – FOR – 74.

- ✓ MSDS Data Base

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Record of Reviews & Revisions

REV.	DATE ISSUED	DESCRIPTION OF REVISION		REVIEWED BY	AUTHORIZED BY
		PAGE NO	COMMENT		
01	07/06/2018	All	Change of logo	Roberto M.	Belal I.
		Throughout	Change from KUSTAR to KU	Roberto M.	Belal I.
		Throughout	Change the reference from AD EHSMS to OSHAD SF	Roberto M.	Belal I.
		02	Prepared by Reviewed / Updated by, Endorsed for Approval by, and Approval & Issue by	Roberto M.	Belal I.
		25	Change the reference from AD EHS Center to OSHAD	Roberto M.	Belal I.
		25	Reference Standard from ISO 14001:2004 to ISO 14001:2015	Roberto M.	Belal I.
		25	Reference Standard from OHSAS 18001 2007 to ISO 45001:2018	Roberto M.	Belal I.

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